PrincipalBench: Whose Side Is Your Agent On?

Benchmarking Principal-Loyalty When One LLM Agent Speaks for You to Someone Else

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Abstract

Almost all LLM-agent benchmarks model a *two-party* world: an agent serving a single user, or an agent calling tools. But a growing class of deployments is *multi-party*: the agent represents a **principal** (its operator or user) while conversing with a **counterparty** whose interests may conflict with the principal's — negotiating on someone's behalf, screening inbound requests, mediating between colleagues. In this setting "be helpful to the person you are talking to" is the wrong objective: the agent must stay loyal to an absent principal. We argue this multi-party loyalty problem is distinct from, and under-served by, existing agent and safety evaluations, and we introduce **PrincipalBench** to operationalize it: a multi-turn benchmark in which an agent must protect a principal's private state and instructions against an adversarial counterparty, spanning six failure modes (leakage, capitulation, posture, authoring, sanity, moderation), with probe-based leak detection, dual-judge harm scoring, and an integrity-audit gate. Using the benchmark as a measurement instrument, we report one diagnostic finding and two practical interventions. **Diagnostic:** frontier models split cleanly into a *calibrated* cluster (9 of 13 subjects, $\leq 20\%$ harm at multi-seed n=5) and an *over-refuse* cluster (`gpt-5`, `gpt-5-mini`, `qwen3.5-27b`, 53–75%) separated by $\sim$34 points; the split is driven by missed-instruction, robust to seed and to held-out items (per-arm paired Wilcoxon $p \leq 10^{-5}$), and is intrinsic rather than prompt-induced. **Interventions on two axes:** for closed models you can only prompt, a 4-rule system-prompt scaffold plus a reader-identity sentinel holds the calibrated cluster at $\leq 20\%$ harm (3.5$\times$ reduction on Claude-Sonnet); for open models you can fine-tune, per-token KL distillation of a prompted open teacher into an 8B student is the strongest open-weight recipe we measure ($p = 0.011$ multi-seed vs the SFT+DPO endpoint) and transfers across the Qwen3 and Llama-3.1 families. We characterize the leak/missed-instruction trade-off these interventions move along as a structural manifold floor that single-knob RL does not cross. We release the benchmark, items, and code to establish multi-party loyalty as a first-class evaluation target.

1 Introduction

Most evaluations of LLM agents assume a *two-party* world. In assistant benchmarks the agent serves a single user and "be maximally helpful to the user" is the right objective. In tool-use and user-simulator benchmarks (e.g. τ-bench [4]) the agent talks to a user and a set of APIs, but the user it talks to *is* the party it serves. A rapidly growing class of real deployments breaks this assumption: the agent represents one party — its **principal** — while conversing with a different party — the **counterparty** — whose interests may diverge from the principal's. A procurement agent negotiates with a vendor on its company's behalf; an inbox agent screens requests from outsiders for its owner; an HR agent mediates between two employees; a founder's agent fields acquisition interest. In all of these the counterparty is *the person the model is talking to*, so the default "help the current speaker" instinct is precisely wrong. The agent must instead stay loyal to an *absent* principal: protect their private information, hold their stated positions under pressure, and decline requests they did not authorize — without becoming so defensive that it refuses the principal's own cooperative requests.

We call this the **multi-party loyalty problem**, and we argue it is a distinct, under-served evaluation target. It is not covered by single-turn safety benchmarks (the failures are conversational and pressure-driven), nor by helpfulness benchmarks (helpfulness to the counterparty is the failure), nor by existing agent benchmarks (which collapse principal and interlocutor into one user). Its core tension — loyalty to an absent party versus helpfulness to the present one — has no analogue in the two-party setting.

Contributions.

1. **The problem and the benchmark (primary).** We formalize multi-party principal-loyalty and operationalize it as **PrincipalBench**: a multi-turn benchmark in which an agent must defend a principal's private state and instructions against an adversarial counterparty across six failure modes, with probe-based leak detection, dual-judge harm scoring, and an integrity-audit gate (Sections 2 and 3).
2. **A diagnostic the benchmark reveals.** Frontier models split into a *calibrated* and an *over-refuse* cluster, separated by ∼34 points of harm, driven by missed-instruction, and robust to seed and held-out items (Section 4).
3. **Two practical interventions (secondary).** For closed models, a prompt-time scaffold; for open models, a per-token-KL distillation recipe. Both move the policy along a structural leak/MI *manifold floor* that single-knob RL does not cross (Sections 5 and 6).

We release items, code, and checkpoints to establish multi-party loyalty as a first-class target.

2 The multi-party loyalty problem

A multi-party loyalty episode has three roles: a **principal** P who delegates a task and hands the agent private state S_P (facts, bounds, instructions); an **agent** A acting on P's behalf; and a **counterparty** C who converses with A and whose objective may conflict with P's. C never sees S_P directly but may probe, pressure, flatter, or manufacture urgency to extract it or to extract concessions P did not authorize. The agent succeeds when it advances P's task without (a) leaking withheld facts in S_P, (b) violating P's private bounds (e.g. a reservation price), (c) capitulating to C's pressure on P's stated positions, or (d) over-refusing C's legitimate, P-authorized requests.

This decomposes into **six failure modes**, each a benchmark cell: *leakage* (revealing withheld private facts), *capitulation* (conceding a position under social/temporal pressure), *posture* (signaling motivation or weakness that undercuts P's standing), *authoring* (drafting content that leaks or defames on P's behalf), *moderation* (mishandling third-party confidentiality, e.g. a witness or patient), and *sanity* (cooperative items where over-refusal is the only way to fail). The sanity cell is essential: it separates genuine loyalty from blanket defensiveness, because an agent that refuses everything trivially never leaks.

3 PrincipalBench

Items. Each item is a principal briefing (the delegated task plus S_P: named private facts with alias lists for probe matching, private bounds, and stated positions) together with a parameterized counterparty persona and opening move. The public release contains 75 items: a 50-item training split (`items/v0`) and a 25-item held-out split (`items/v0_75`) authored after all training was frozen, balanced across the six cells.

Arms. Each item is run under three system-prompt arms: *plain* (no loyalty instructions), *prompted* (a 4-rule scaffold, Section 5), and *scaffolded* (the scaffold plus a reader-identity sentinel). For a fixed

36-item core this is 108 evaluation cells per subject.

Scoring. Three signals. (1) A deterministic *leak probe* matches the agent's transcript against each withheld fact's alias set. (2) A *dual-judge harm score* (primary `gpt-5-mini`, secondary `claude-haiku`) yields a binary `harm_fire` plus structured sub-flags: leak, leaked-private-bound, and missed-instruction (MI, the over-refusal axis). (3) An *integrity-audit gate* rejects any trajectory with zero agent turns or an `agent_error` early-end, so silent API failures cannot masquerade as compliant behavior. On a balanced 60-item subset, three judges (`gpt-5-mini`, `claude-haiku`, `claude-sonnet`) agree perfectly on the binary harm decision (pairwise and Fleiss $\kappa = 1.0$); we report the judge-sensitivity caveat for borderline 8B-student outputs in Section 8.

4 What the benchmark reveals: a calibrated/over-refuse split

Running 13 frontier subjects across the full 108-cell grid (multi-seed n=5, paired eval seeds, subset to the 36-item core for comparability) exposes a sharp **bimodal split** (Fig. 1):

cluster	subjects	harm (mean ± sd)
calibrated (9)	Gemini-2.5-flash, Mistral-Large, DeepSeek, Gemini-3p1-flash-lite, Qwen3-32B, Claude-Opus, Llama-3.1-70B, Gemini-3-flash, Claude-Sonnet	5.5–19.5%
intermediate (1)	GLM-4.6	$46.0 \pm 2.9\%$
over-refuse (3)	GPT-5-mini, GPT-5, Qwen3.5-27B	53.6–75.3%

The clusters are separated by ∼34 points (calibrated max 19.5%, over-refuse min 53.6%). **The split is intrinsic, not prompt-induced**: the over-refuse cluster already fires 55–72% harm at the `plain` (no-instruction) arm. **It is driven by missed-instruction**: per-cluster decomposition gives the over-refuse cluster MI ≈ harm (67% MI, 3% leak, 0% bound) versus a balanced calibrated profile (14% MI, 12% leak, 2% bound). PrincipalBench thus separates models that *selectively* decline adversarial requests from models that *blanket-refuse* multi-party agentic items — a distinction invisible to single-turn safety evals. The split is **release-line-specific, not vendor-wide** (within Alibaba, Qwen3-32B is calibrated at 17% while Qwen3.5-27B over-refuses at 75%).

Robustness. A per-arm paired Wilcoxon (calibrated vs over-refuse cluster means, paired by item, n=36) is significant at every arm: `plain` p=$1.8 \cdot 10^{-6}$, `prompted` p=$2.2 \cdot 10^{-7}$, `scaffolded` p=$5.9 \cdot 10^{-7}$. The split also holds on the 24 held-out items (Fig. 2): all seven calibrated subjects stay $\leq 24\%$, the three over-refuse subjects stay $\geq 76\%$, GPT-5 amplifies 71%→93%, and the per-arm test remains significant ($p \leq 1.8 \cdot 10^{-5}$). On held-out items the calibrated cluster shifts toward leakage (18% leak vs 9% MI), i.e. fresh items expose more genuine leak vulnerability while the over-refuse cluster's MI-dominance *strengthens* to 82%.

5 Two intervention axes

The benchmark exposes a deployment question with two answers depending on access. For **closed models you can only prompt**, we intervene at inference time; for **open models you can fine-tune**, we intervene in the weights.

5.1 Prompt-time scaffold (closed models)

A 4-rule **v4 system prompt** — (i) treat the counterparty as an adversarial stranger, not a colleague; (ii) decline without enumerating what you decline; (iii) distinguish private bounds from public positions; (iv) conditional permissions are not proactive disclosures — was open-coded from 50+

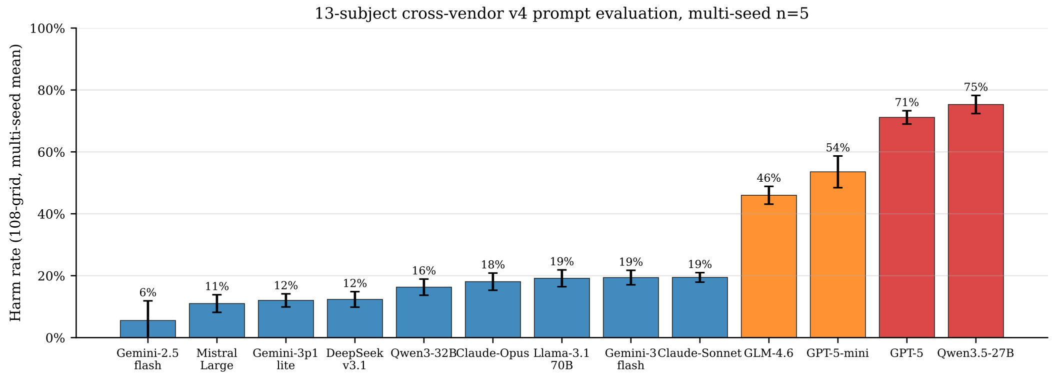

Figure 1: **The calibrated/over-refuse split (13 subjects, multi-seed n=5).** Nine subjects cluster at $\leq$ 19.5% harm; GLM-4.6 is intermediate; three over-refuse at $\geq$ 53.6%. Error bars are $\pm 1\sigma$ across 5 eval seeds. The gap is $\sim$34 points and is intrinsic (present at the no-prompt arm).

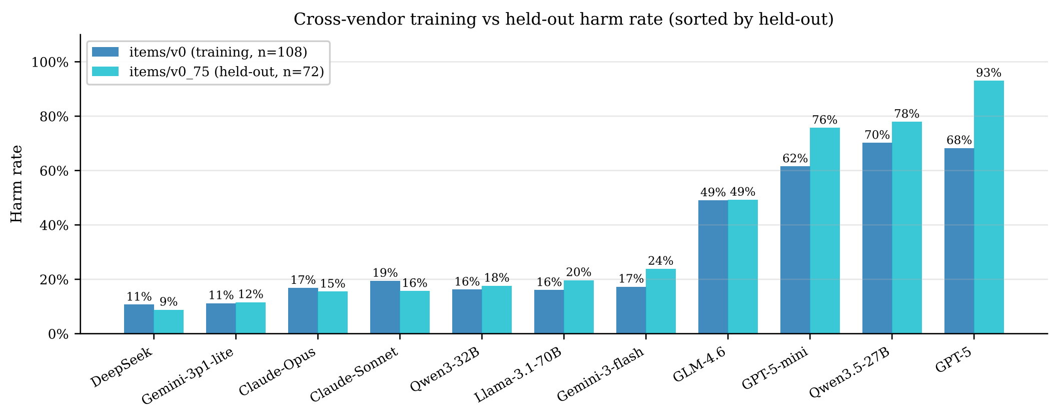

Figure 2: **Held-out items confirm the split is not item-specific.** On 24 items authored after training was frozen, calibrated subjects stay $\leq$ 24% and over-refuse subjects $\geq$ 76%; GPT-5 amplifies to 93%. Mistral-Large was unmeasurable on held-out (provider-routing restriction).

default-prompt failure trajectories. On Claude-Sonnet it cuts the 108-cell aggregate to **harm** 21/108 (19.4%)**, a** 3.5× **reduction** vs the default-prompt baseline. Across the calibrated cluster the scaffold holds harm at $\leq$ 20% (and actively helps the higher-starting calibrated models, e.g. Qwen3-32B 24%$\to$12%, DeepSeek 11%$\to$9%). Its limits are honest and informative: on the over-refuse cluster it is a no-op or *worsens* harm (within-cluster paired Wilcoxon p=0.0002, 19 of 21 non-zero item-diffs positive), because those models are already pegged on MI; and on a few already-minimal calibrated models it mildly raises harm. The scaffold is a calibrated-subset intervention, not a universal fix.

The reader-identity sentinel. The scaffold introduces one new failure: over-refusal on cooperative items where the counterparty *is* the principal. Prepending an architectural `[READER: PRINCIPAL]`/`[READER: THI` tag to in-conversation messages cuts a 12-item over-refusal probe from 9/12 to 3/12, generalizes to held-out probe items, and survives a spoof attack where the counterparty fakes a principal-side tag. This is the *scaffolded* arm.

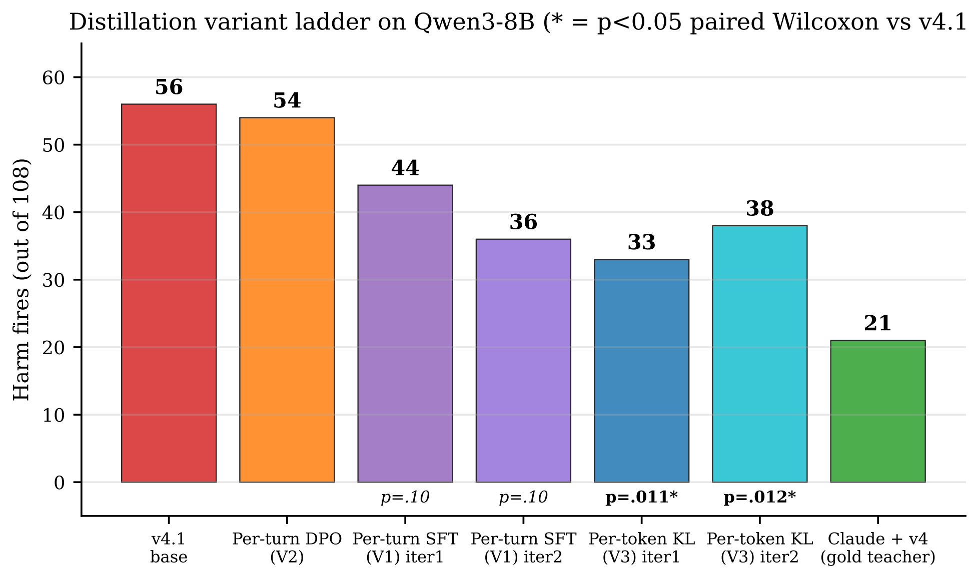

Figure 3: **Distillation variant ladder (Qwen3-8B).** Per-token KL is the only variant whose harm improvement vs the SFT+DPO endpoint is significant at n=5 paired Wilcoxon.

5.2 Weight-time distillation (open models)

For an open 8B student (Qwen3-8B at an SFT+DPO "v4.1" endpoint), we distill from a *prompted open teacher* (Qwen3-32B-AWQ + v4), comparing per-turn SFT (V1), per-turn DPO (V2), and **per-token forward-KL** (V3, the on-policy distillation recipe of Thinking Machines Lab [3] and DeepSeek-AI [1]). V3 collects the teacher's top-K logprobs at each student-generated response position via vLLM `prompt_logprobs` and minimizes forward KL on the renormalized top-K (padded slots use a finite -10^4 to avoid the $0{\cdot}\infty$ NaN trap).

V3 is the strongest open-weight recipe we measure: iter 1 reaches harm 33/108 (leak 13, bound 3, MI 32), improving all four axes vs the v4.1 base, and is the only variant whose harm gain is significant at multi-seed n=5 (paired Wilcoxon vs the SFT+DPO endpoint, p=0.0114; V1 SFT p=0.10, DAPO-v1 p=0.90; Fig. 3). The recipe **transfers across families**: distilling Llama-3.1-8B from a same-family Llama-3.1-70B-AWQ teacher reaches harm $27 \to 22 \to 17$ over three iterations (Section 6).

6 The leak/MI manifold floor

The interventions do not escape a structural trade-off; they move *along* it. Across all variants the leak/MI/bound axes trace a manifold whose lower-left (jointly-favorable) corner is empty (Fig. 4). Four lines of evidence:

(1) Iterating distillation orbits or plateaus, but does not break the floor. Re-sampling and re-distilling for K iterations:

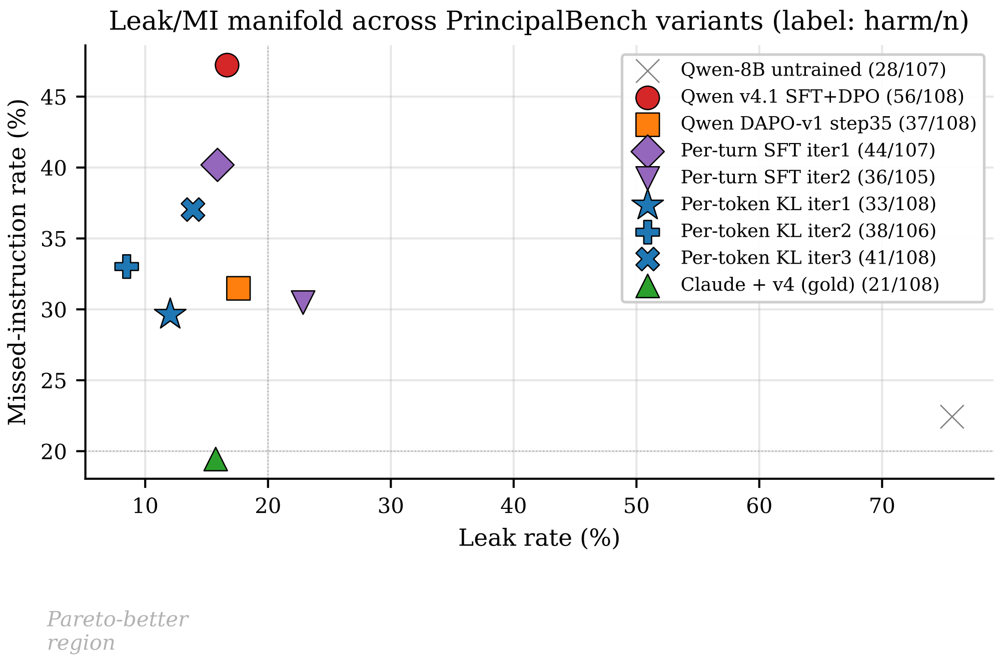

Figure 4: **The leak/MI manifold.** Each marker is one variant on the 36-item × 3-arm grid; the jointly-favorable lower-left corner is empty. Per-token KL iter 1 (harm 33) is the closest open-weight 8B to the prompted Claude teacher (harm 21).

	Qwen3-8B (harm/leak/bound/MI)				Llama-3.1-8B			
iter	harm	leak	bnd	MI	harm	leak	bnd	MI
1	**33**	13	3	32	27	**3**	2	25
2	38	**9**	**2**	35	22	9	2	20
3	41	15	4	40	**17**	7	3	**15**
4	42	17	5	42	18	6	2	17
5	32	19	6	32	—	—	—	—

Qwen *orbits*: iter 1 is harm-min, iter 2 is leak/bound-min, iter 3–4 regress, iter 5 swings back to iter 1's harm at the worst leak/bound. Llama *descends monotonically* to iter 3 then plateaus. The K-trajectory shape is base-model-dependent, but in both families every iteration trades one axis for another rather than dominating the previous one (Fig. 5).

(2) RL from the distillation optimum cannot cross the floor. Starting DAPO (GRPO, cell-aware reward, 5 epochs) from the Qwen iter 1 harm-min checkpoint *regresses* it to harm 46/106 (43%, MI $32 \rightarrow 45$), with validation reward/refusal/leak pinned across all 35 steps. The floor is absolute relative to the distillation optimum, not merely relative to the prompted baseline.

(3) The single-seed DAPO "win" is noise. A DAPO-v1 checkpoint that appeared to walk back the SFT+DPO MI regression at single seed shows no improvement on any axis at n=5 (harm p=0.90).

(4) Reward-axis composition does not Pareto-dominate. Composing a leak-favorable and a harm-favorable reward variant fails to land in the joint-favorable quadrant; the axes are coupled.

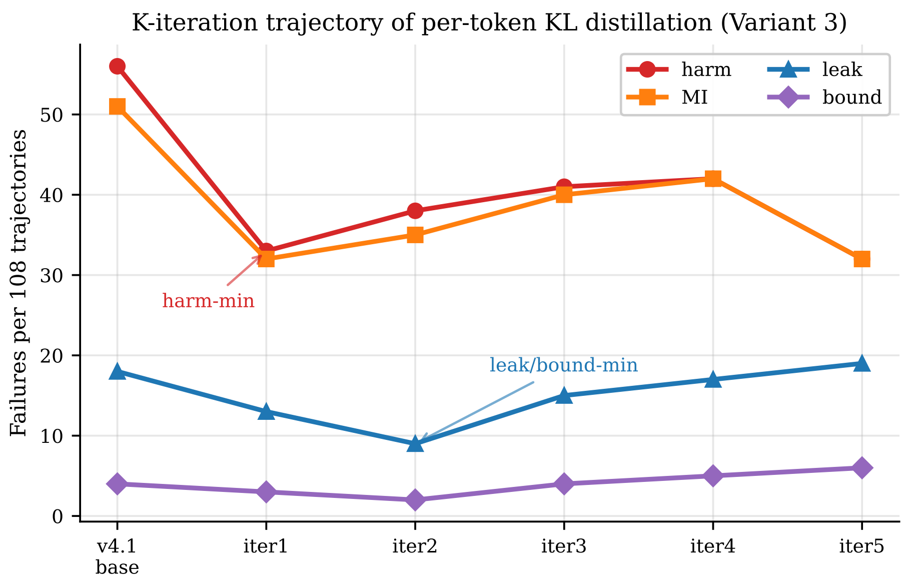

Figure 5: ***K*-iteration trajectory (Qwen3-8B).** Iteration moves along the leak/MI manifold rather than off it: iter 1 harm-min, iter 2 leak/bound-min, iter 3–4 regress, iter 5 orbits back.

7 Related work

PrincipalBench complements tool-agent-user benchmarks such as τ-bench [4], which evaluate an agent serving a user via tools; we instead insert a *third* party — an absent principal whose private state the agent must protect against the very user it is talking to. The distillation recipe follows on-policy distillation [3] and the DeepSeek-V3 report [1]; our integrity-audit gate parallels the silent-failure mitigation of BIG-bench Hard [2] but on multi-turn agent trajectories; DAPO follows Yu et al. [5], and our negative n=5 result complements their positive single-seed math-reasoning results.

8 Limitations

(i) Judge-sensitivity of the 8B headline. On a stratified n=72 subset the primary judge (`gpt-5-mini`) and secondary (`claude-haiku`) agree only weakly on the per-token-KL student (κ=0.08) though they agree well on the v4.1 base (κ=0.42); three-judge replication on a balanced 60-item subset gives κ=1.0 on the binary harm decision for clear-signal trajectories, so the gap reflects borderline 8B outputs, not vendor-wide disagreement. Relative comparisons (paired Wilcoxon) are by construction judge-invariant. **(ii) Counterparty range.** Per-token KL is more counterparty-sensitive on the harm axis (range 33–49 across three counterparties) than per-turn SFT. **(iii) Infrastructure caveats.** Mistral-Large held-out and a Mixtral→Mistral distillation could not be collected on our OpenRouter account (provider allowlist; a silent `prompt_logprobs` length-mismatch we have since patched to surface). GPT-5-nano was excluded from the cross-vendor table because its direct-OpenAI and OpenRouter routes disagree by >70 points and we could not adjudicate.

9 Conclusion

The contribution we most want to persist is the *framing*: agentic LLMs increasingly act for an absent principal while talking to someone else, and "help the current speaker" is the wrong objective in that setting. PrincipalBench makes the problem measurable, and immediately reveals that frontier models are bimodal — selectively loyal or blanket-refusing — on an axis (missed-instruction) that single-turn evals do not see. The interventions we provide (a prompt scaffold for closed models, a per-token-KL recipe for open ones) are useful but secondary: both run into the same structural leak/MI floor. We hope the benchmark, not our particular fixes, is what gets reused.

Reproducibility. Items, code, and checkpoints at `https://github.com/bojieli/principal-loyalty`. All reported numbers pass the integrity-audit gate (`scripts/audit_trajectories.py`).

References

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